The C++ Standard Template Library

Containers hold data, algorithms work on it, iterators connect them. Container choice, operation costs and iterator invalidation are the parts that matter most.

The big picture



Algorithms never touch containers, only iterator ranges.

One sort works on any container that offers the right iterators.

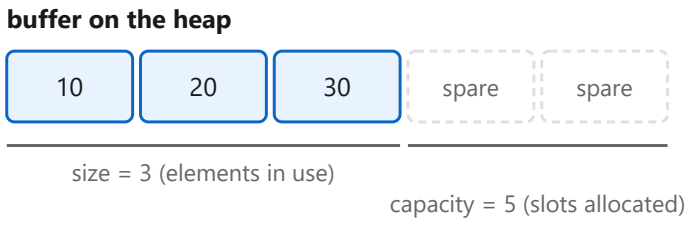
PART	WHAT IT IS	EXAMPLES
Containers	classes that own and organise a collection of elements	`vector`, `map`, `set`, `deque`
Iterators	generalised pointers that walk a container	`v.begin()`, `m.find(k)`
Algorithms	generic functions that operate on iterator *ranges*	`sort`, `find`, `accumulate`

Algorithms see only `[first, last)`, so one `std::sort` serves a vector, array or deque. That separation is the design.

```
#include <vector>       // std::vector
#include <map>          // std::map, std::multimap
#include <unordered_map>// std::unordered_map
#include <set>          // std::set
#include <algorithm>    // std::sort, std::find, std::transform ...
#include <numeric>      // std::accumulate, std::iota
#include <string>       // std::string
```

DEFAULT CHOICE `std::vector`. Contiguous memory is so cache-friendly it often beats "better" containers.

`vector`: the workhorse



push_back, buffer full:
allocate bigger, move all, free old
= amortised O(1), but every iterator and pointer into it is invalidated

reserve() grows capacity only; resize() changes size.

A `vector` is one contiguous heap array with a **size** and a **capacity**. When full, `push_back` allocates a bigger buffer, moves everything and frees the old one: occasional O(n), **amortised O(1)**.

```
std::vector<int> v;
// capacity 100, size still 0: no reallocation for the next 100 push_backs
v.reserve(100);
v.push_back(1);          // copy/move an existing object in
v.emplace_back(2);       // construct the element in place: no temporary
// size becomes 10; new elements are value-initialised (0)
v.resize(10);

// no bounds check: out-of-range is undefined behavior
v[3];
v.at(3);                 // bounds-checked: throws std::out_of_range
v.front(); v.back(); v.data();   // data() is the raw int* for C APIs
v.size(); v.capacity(); v.empty();
```

RESERVE VS RESIZE `reserve(n)` only allocates; `resize(n)` creates elements. `v[5] = x` after only `reserve(10)` is UB.

BRACES VS PARENTHESES

```
// parentheses: size 3, every element 7 → {7, 7, 7}
std::vector<int> a(3, 7);
// braces: the initializer_list constructor wins → {3, 7}
std::vector<int> b{3, 7};
```

With braces, a constructor taking `std::initializer_list` is preferred whenever it can match.

Iterators

An iterator acts like a pointer: `*it`, `++it`, `it == end`. `end()` is **one past the last**; ranges are half-open `[begin, end)`.

```
std::vector<int> v{1, 2, 3};
// use ≠ not <, it works for every container
for (auto it = v.begin(); it ≠ v.end(); ++it) std::cout << *it;
// range-for is exactly this loop in disguise
for (int x : v) std::cout << x;
// reverse iterators
for (auto it = v.rbegin(); it ≠ v.rend(); ++it) { }
// cbegin/cend: read-only, *it cannot be modified
for (auto it = v.cbegin(); it ≠ v.cend(); ++it) { }
// advance without assuming random access
auto it2 = std::next(v.begin(), 2);
```

CATEGORY	CAN DO	CONTAINERS
Forward	`++`, read (multi-pass)	`forward_list`, unordered containers
Bidirectional	also `--`	`list`, `map`, `set`
Random access	also `it + n`, `it[i]`, `it1 - it2`, `<`	`vector`, `deque`, `array`, `string`

So `std::sort` needs random access and fails on `std::list` (use `list.sort()`).

Iterator adaptors: filling a container

An algorithm like `std::copy` **assigns** through its output iterator, so the destination must already be big enough. An **inserter** turns each assignment into an insertion instead:

ADAPTOR	EACH ASSIGNMENT DOES
`std::back_inserter(c)`	`c.push_back(x)`
`std::front_inserter(c)`	`c.push_front(x)` (so the order reverses; needs `deque` / `list`)
`std::inserter(c, pos)`	`c.insert(pos, x)` (works for `set` and `map`)

```
std::vector<int> src{3, 1, 2}, out;
// out is {3, 1, 2}: it grew
std::copy(src.begin(), src.end(), std::back_inserter(out));
// std::copy(src.begin(), src.end(), out.begin()); // BUG if out is empty:
// writes past the end

std::set<int> sorted;
std::copy(src.begin(), src.end(), std::inserter(sorted, sorted.begin()));
```

`std::reverse_iterator` (what `rbegin()` returns) and `std::move_iterator` (moves instead of copying) are the other two adaptors worth knowing.

Iterator invalidation

Changing a container can **invalidate** iterators, pointers and references; using one is UB.

CONTAINER	OPERATION	WHAT IS INVALIDATED
`vector`	`push_back`, `insert`	**all** if it reallocated; otherwise those at or after the insertion point
`vector`	`erase`	the erased element and everything after it
`deque`	insert/erase in the middle	everything; push at either end invalidates iterators but not references
`list`, `map`, `set`	insert	nothing
`list`, `map`, `set`	erase	only the erased element
`unordered_*`	insert that triggers a rehash	all iterators (references stay valid)

▼ SPOT THE BUG **Erasing inside a loop**

```cpp
std::vector<int> v{1, 2, 3, 4};
for (auto it = v.begin(); it != v.end(); ++it)
    if (*it % 2 == 0) v.erase(it);          // BUG
```

`erase` invalidates `it`, and the loop then does `++it` on a dead iterator. The correct idiom uses the iterator `erase` returns:

```cpp
for (auto it = v.begin(); it != v.end(); ) {
    // erase returns the next valid iterator
    if (*it % 2 == 0) it = v.erase(it);
    else ++it;
}
// or, in C++20:
std::erase_if(v, [](int x) { return x % 2 == 0; });
```

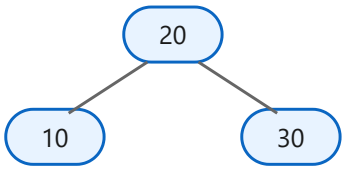
Other sequence containers

CONTAINER	STORAGE	BEST AT	WATCH OUT
`std::array<T,N>`	fixed size, inline (stack)	size known at compile time; zero overhead	cannot grow
`std::vector<T>`	one contiguous heap block	almost everything; fast iteration and indexing	insert/erase in the middle is O(n)
`std::deque<T>`	chunks of arrays	fast push/pop at **both** ends, still indexable	not contiguous
`std::list<T>`	doubly linked nodes	O(1) insert/erase anywhere *if you hold an iterator*; iterators stay valid	no indexing, a heap allocation per element, poor cache behaviour
`std::forward_list<T>`	singly linked nodes	minimal memory list	forward iteration only

OPERATION	VECTOR	DEQUE	LIST
index `[i]`	O(1)	O(1)	not available
`push_back`	amortised O(1)	O(1)	O(1)
`push_front`	O(n)	O(1)	O(1)
insert / erase in the middle	O(n)	O(n)	O(1) at a known position
find a value	O(n)	O(n)	O(n)

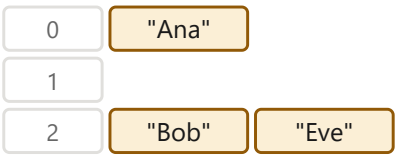
Associative and unordered containers

map / set: balanced tree, sorted, O(log n)



in-order walk = sorted order

unordered_map: hash table, average O(1)



hash(key) picks a bucket; collisions chain (Bob, Eve)

Need order or range queries: map. Plain lookup: unordered_map.

	`MAP` / `SET`	`UNORDERED_MAP` / `UNORDERED_SET`
Implementation	balanced binary tree (usually red-black)	hash table
Order of iteration	**sorted** by key	unspecified
find / insert / erase	O(log n)	average O(1), worst case O(n)
Key requirement	`operator<` (a strict weak ordering)	a hash function and `operator==`
Range queries (`lower_bound`)	yes	no

```cpp
std::map<std::string, int> age;
// operator[] INSERTS a default (0) if the key is missing
age["Ana"] = 30;
// does nothing if "Bob" already exists
age.insert({"Bob", 25});
age.insert_or_assign("Bob", 26);          // C++17: insert or overwrite

if (auto it = age.find("Ana"); it != age.end())   // find never inserts
    // it->first is the key, it->second the value
    std::cout << it->second;

// structured bindings; sorted by name
for (const auto& [name, years] : age)
    std::cout << name << ' ' << years << '\n';
```

```cpp
std::unordered_set<int> seen;
if (!seen.insert(5).second) { /* second == false means it was already there
*/ }
```

`try_emplace`, `multimap` and `multiset`

CALL	IF THE KEY ALREADY EXISTS
`m[k] = v`	overwrites (and inserts a default first if missing)
`m.insert({k, v})` / `m.emplace(k, v)`	does nothing, but **may already have moved from or built the value**
`m.try_emplace(k, args ... )` (C++17)	does nothing and **guarantees the arguments are not touched**: compiled, a moved-in 40-character string was intact afterwards
`m.insert_or_assign(k, v)` (C++17)	overwrites

```cpp
std::map<int, std::string> a{{1, "old"}};
std::string payload(40, 'p');
// key 1 exists: nothing happens, payload still holds 40 chars
a.try_emplace(1, std::move(payload));
// new key: the string is moved into the map
a.try_emplace(2, std::move(payload));

// duplicate keys allowed
std::multimap<std::string, int> mm{{"fruit", 1}, {"fruit", 2}, {"veg", 9}};
// the range of all entries with that key
auto [lo, hi] = mm.equal_range("fruit");
// 2: there is no operator[] because a key may have many values
mm.count("fruit");
```

- `multimap` and `multiset` allow duplicate keys and keep equal keys in insertion order. There is no `operator[]`.
- `try_emplace` is the right call for "insert if absent" when building the value is expensive or the argument is move-only.

Container adaptors

They wrap another container and restrict the interface: `std::stack` (LIFO: `push/pop/top`), `std::queue` (FIFO: `push/pop/front`) and `std::priority_queue` (a heap: `top()` is the **largest** element by default).

```
// largest first
std::priority_queue<int> maxHeap;
// smallest first
std::priority_queue<int, std::vector<int>, std::greater<int>> minHeap;
```

Which container should I use?

I NEED TO...	USE	WHY
iterate over items, access by index, mostly append at the end	`std::vector`	Contiguous memory: fastest iteration and indexing, amortised O(1) `push_back`. This is the default choice.
hold a fixed number of items known at compile time	`std::array`	Lives inline (no heap), knows its size, zero overhead over a C array.
push and pop at both ends (a deque of work)	`std::deque`	O(1) push/pop at front and back while still allowing indexing.
insert/erase often in the middle while holding iterators that must stay valid	`std::list`	O(1) splice/insert/erase at a known position and stable iterators. Costs: a heap allocation per node and poor cache locality. Measure before choosing it.
look up values by key; order does not matter	`std::unordered_map`	Hash table: average O(1) find/insert/erase. Needs a hash and `operator==`.
look up by key AND iterate in sorted order, or find the nearest key	`std::map`	Balanced tree: O(log n), sorted iteration, `lower_bound` / `upper_bound` for range queries.
check membership / remove duplicates (no values attached)	`std::unordered_set`	Hash set: average O(1) contains. Use `std::set` if you also need the elements sorted.
count how often each word occurs	`std::unordered_map<std::string, int>`	`counts[word]++` inserts a zero on first sight and increments in one expression.
last-in first-out (undo, DFS, matching brackets)	`std::stack`	Adaptor over deque/vector exposing only push, pop and top.
first-in first-out (task queue, BFS)	`std::queue`	Adaptor over deque exposing push, pop, front and back.
always process the largest (or smallest) item next	`std::priority_queue`	A binary heap: O(log n) push/pop, O(1) top. Largest first by default; pass `std::greater` for smallest first.

KEY POINT "vector vs list?" Answer with *memory*: contiguous means cache-friendly, so vector wins even for middle inserts at modest sizes. List wins for stable iterators and cheap splicing.

Custom keys: comparators and hashes

A `map` / `set` needs an **ordering**; an `unordered_map` / `unordered_set` needs a **hash and equality**. For your own key types, you provide them.

```
// an ordering that ignores case
struct NoCase {
    bool operator()(const std::string& a, const std::string& b) const {
        return std::lexicographical_compare(a.begin(), a.end(), b.begin(),
b.end(),
            [](unsigned char x, unsigned char y) { return std::tolower(x) <
std::tolower(y); });
    }
};
// the comparator is part of the TYPE
std::map<std::string, int, NoCase> m;
// the same key: size is 1 (compiled: 2 keys with "beta")
m["Alpha"] = 1; m["ALPHA"] = 2;

struct Product { std::string shop; int id; bool operator==(const Product&)
const = default; };
struct ProductHash {
    std::size_t operator()(const Product& p) const {
        std::size_t h = std::hash<std::string>{}(p.shop);
        // combine the member hashes
        h ^= std::hash<int>{}(p.id) + 0x9e3779b97f4a7c15ULL + (h << 6) + (h
>> 2);
        return h;
    }
};
std::unordered_map<Product, double, ProductHash> price;
price[{"north", 7}] = 101.5;
```

CONTAINER	KEY MUST PROVIDE	RULE
`map`, `set`	a **strict weak ordering** (`operator<` or a comparator)	irreflexive and transitive; equivalence means "neither is less". `≤` is **not** valid: undefined behaviour
`unordered_map`, `unordered_set`	a hash and `operator==`	equal keys **must** have equal hashes; a poor hash makes every lookup O(n)

- **Transparent comparators:** `std::map<std::string, int, std::less<>>` lets `find` take a `string_view` or `const char*` without building a temporary `std::string` (compiled: it found `"apple"` from a `string_view`). C++20 adds the same for unordered containers.
- **Never change a key** while it is in the container: its position or bucket would be wrong. Keys are `const` for that reason.
- **Hash quality:** combine *all* fields; avoid returning just one field. `std::hash` for `int` is the identity in libstdc++, which is fine.
- A `rehash` (growth beyond `max_load_factor`) invalidates iterators; `reserve(n)` up front avoids it, and avoids the allocations on a hot path.
- Ordered or hashed? Ordered when you need sorted iteration or range queries (`lower_bound`); hashed for plain lookup. For a few dozen keys a sorted `vector` and binary search often beats both.

KEY POINT "How do you use your own type as a key in `unordered_map`?" Provide `operator==` and a hash functor (or specialise `std::hash`), combining all members, and make sure equal objects hash equally.

Algorithms

Over a hundred generic functions in `<algorithm>` / `<numeric>`. Prefer them to loops: named, tested, often faster. Most take a lambda.

```
// custom order
std::sort(v.begin(), v.end(), [](const P& a, const P& b) { return a.age <
b.age; });
auto it = std::find_if(v.begin(), v.end(), [](int x) { return x > 6; });
int n  = std::count_if(v.begin(), v.end(), [](int x) { return x % 2 == 0; });
bool any = std::any_of(v.begin(), v.end(), [](int x) { return x < 0; });
int sum  = std::accumulate(v.begin(), v.end(), 0);          // <numeric>
std::transform(v.begin(), v.end(), v.begin(), [](int x) { return x * x; });
// binary search: range MUST be sorted
auto pos = std::lower_bound(sorted.begin(), sorted.end(), 5);
```

> **STRICT WEAK ORDERING** Use `<`, never `≤`: a comparator true for equal elements can crash `sort` or corrupt a `set`. `sort` is not stable; use `stable_sort`.

Ranges and views (C++20)

```
#include <ranges>

// pass the container, not begin() and end()
std::ranges::sort(v);
// descending (compiled: first element 6)
std::ranges::sort(v, std::greater<>{});

auto evenSquares = v | std::views::filter([](int x) { return x % 2 == 0; })
                     // a LAZY view: nothing is computed yet
                     | std::views::transform([](int x) { return x * x; });
// computed on demand, no temporary vector (compiled: sum 56)
for (int x : evenSquares) use(x);
```

- A **range** is anything with `begin()` and `end()`. The `std::ranges::` algorithms take the whole range and support **projections**: `std::ranges::sort(people, {}, &Person::age)`.

- A **view** is a lightweight, non-owning, lazy window (`filter`, `transform`, `take`, `drop`, `reverse`, `iota`). Pipe them with `|`.
- A view does not own its data: if the container dies or reallocates, the view dangles, like `string_view`.
- Ranges make pipelines readable; you do not need to know every view.

Common pitfalls

`pair`, `tuple`, `optional` and `variant` are covered in Vocabulary Types.

PITFALL	FIX
Using an iterator/pointer/reference after a `push_back` reallocated the vector	`reserve` up front, or re-fetch the iterator, or index instead
`map[key]` silently inserts	use `find` / `contains` to test; `at()` to read-or-throw
`v.size() - 1` on an empty vector (`size()` is unsigned, so it wraps to a huge number)	check `empty()` first or write `i + 1 < v.size()`
Passing a big container by value	pass `const std::vector<T>&`
`std::vector<bool>` is a packed-bit special case with proxy references	use `std::vector<char>` or `std::bitset`
`auto x = v[0]` copies the element	`const auto& x = v[0]`
`std::find` on a `map` is O(n)	use the member `m.find(key)`, O(log n)

Random numbers: `<random>`

```cpp
// an ENGINE: produces raw random bits (here seeded for repeatability)
std::mt19937 gen(12345);
// a DISTRIBUTION: shapes the bits into the numbers you want
std::uniform_int_distribution<int> dice(1, 6);
int roll = dice(gen);

// a non-deterministic seed source (hardware or OS)
std::random_device rd;
std::mt19937 gen2(rd());
std::normal_distribution<double> noise(0.0, 1.0);
```

- **Engine plus distribution**: never use `rand() % n` (poor quality, biased, global state). Same seed gives the **same sequence** (compiled: true), which is what you want in tests and simulations.
- Compiled: 6000 rolls of the dice gave every face between 850 and 1150 times; 10,000 normal samples had a mean near 0.
- Keep one engine per thread; engines are not thread-safe.

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
vector: the workhorse	C++ Primer	3.3 Library vector type	125 to 134
	C++ Primer	9.3 Sequential container operations	370 to 383
	C++ Primer	9.4 How a vector grows	384 to 388
	Programming: Principles and Practice	3.5 to 3.6 Functions; vector introduced	81 to 89
	Programming: Principles and Practice	17.6 to 17.9 Other operations, remaining Vector problems, changing size	513 to 523
	Effective Modern C++	Item 42, Consider emplacement instead of insertion	310 to 320
	C++ Move Semantics	Chapter 15 Move semantics in standard library types	229 to 250
Iterators	C++ Primer	3.4 Introducing iterators	135 to 141
	C++ Primer	10.4 Revisiting iterators	430 to 438
	C++ Primer	10.5 Structure of generic algorithms	439 to 443
	Programming: Principles and Practice	19.1 to 19.3, 19.6 Containers, sequences and iterators, linked lists, vector/list/string	559 to 572 and 585 to 586
Other sequence containers	C++ Primer	9.1 to 9.2 Overview and the container library	355 to 369
	C++ Primer	9.6 Container adaptors	397 to 402
	C++ Primer	10.6 Container-specific algorithms	444 to 447
Associative and unordered containers	C++ Primer	11.1 to 11.2 Using an associative container, overview	449 to 456
	C++ Primer	11.3 Operations on associative containers	457 to 471
	Programming: Principles and Practice	20.1 to 20.6 map, unordered_map, set, container overview	591 to 609
Which container should I use?	Programming: Principles and Practice	20.1 to 20.6 map, unordered_map, set, container overview	591 to 609

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Custom keys: comparators and hashes	C++ Primer	11.3 Operations on associative containers	457 to 471
	C++ Primer	11.4 The unordered containers	472 to 477
Algorithms	C++ Primer	10.1 to 10.2 Overview and a first look	405 to 413
	Programming: Principles and Practice	21.1 to 21.2, 21.5 Standard algorithms, function objects, sorting and searching	617 to 626 and 633 to 635
	C++ Move Semantics	Chapter 14 Moving algorithms and iterators	219 to 228
Common pitfalls	C++ Primer	9.3 Sequential container operations	370 to 383
Random numbers	C++ Primer	17.4 Random numbers	774 to 780